Deferred Emergence Causality

Canonical Research Artifact (v1.0)

Overview

This repository contains a paired body of work consisting of:

1. A rigorous mathematical forcing framework for the odd Collatz map, identifying an intrinsic delay invariant enforced by exact arithmetic constraints.
2. A generalized causal principle, *Deferred Emergence Causality*, abstracted from that framework and independently observed in AI-like dynamical systems.

Together, these documents identify a causal regime in which transitions are not merely slow or unlikely, but causally forbidden until an intrinsic, irreversible constraint budget is exhausted.

Documents

1. A Finite-Core Forcing Approach to the Collatz Conjecture via 2-Adic Delay Invariants

This paper develops a branch-consistent forcing model for the odd Collatz dynamics at fixed 2-adic precision. It introduces:

- a finite directed forcing graph at each precision,
- exact lifting lemmas controlling behavior under refinement,
- an intrinsic delay invariant,
- provable lower bounds on reachability, and
- a linear growth law for worst-case forcing depth.

The work isolates a structural obstruction: deep valuation transitions are causally unreachable before a minimum number of steps, regardless of strategy.

This document is purely mathematical and does not claim a proof of the Collatz conjecture.

2. Deferred Emergence Causality

This paper extracts and formalizes a domain-agnostic causal principle from the arithmetic framework above and from an independent output-only structural detection system.

The principle states:

> Certain transitions are causally forbidden until intrinsic constraint budgets are exhausted. During this delay phase, structure accumulates invisibly. When the budget is depleted, transition becomes possible and often appears sudden.

The abstraction does not propose a physical theory or new force. It identifies a minimal causal regime based on irreversible constraint ordering, rather than spacetime geometry, energy barriers, or probabilistic waiting.

What Is New Here

This work identifies and formalizes a causal regime in which transitions are not merely slow or improbable, but causally forbidden until an intrinsic, irreversible constraint budget is exhausted. The regime is instantiated exactly in a branch-consistent forcing framework for the odd Collatz map via a valuation-derived delay invariant, and abstracted as Deferred Emergence Causality.

While delay, thresholds, and emergence have been widely discussed, the enforcement of delay as structural impossibility—independent of spacetime geometry, energy barriers, or probabilistic waiting—has not previously been formalized in this way.

The work does not claim a solution to the Collatz conjecture or a physical theory, but isolates a minimal, domain-agnostic causal principle grounded in exact arithmetic dynamics.

Scope and Non-Claims

This work does not claim:

- a solution to the Collatz conjecture,
- a new physical force or field,
- a complete theory of spacetime or intelligence,
- inevitability claims about AI or cosmology.

This work does claim:

- the identification of an intrinsic, unavoidable causal delay enforced by constraint structure,
- a concrete arithmetic instantiation of causality without geometry,
- a general causal pattern applicable across distinct domains.

Authorship and Method

This work was developed through iterative human–AI collaboration, with the human author directing conceptual framing, interpretation, and scope, and the AI system supporting formal exploration, consistency checking, and articulation.

The discovery emerged through dialogue rather than automation.

Versioning

This repository represents the canonical v1.0 articulation of the work.

Subsequent extensions, speculation, or applications—if any—should be treated as separate from this core statement and must not revise the original formulation.

Status

This work is intentionally released without expectation of immediate adoption. It is preserved as a stable scientific artifact for independent evaluation, future reuse, or reinterpretation.

Canonical Version Lock

This release is tagged: `v1.0-canonical`.

The PDFs and this README are preserved without revision.

Any future material must appear only in:

- `/notes/`

- `/extensions/`
- `/reflections/`

Canonical v1.0 research artifact — preserved without revision.